🚀 Installing Kali Linux with Docker Compose

In this session, we'll set up a Kali Linux container using Docker Compose. We'll walk through the configuration and explain why Kali Linux is a great choice for secure computing labs.

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🔐 Why Kali Linux?

- **Security-Focused Distribution:**
  Tailored for penetration testing, ethical hacking, and digital forensics.
- 🛠️ **Comprehensive Toolset:**
  Comes pre-installed with hundreds of security tools.
- 🔄 **Active Community & Regular Updates:**
  Ensures the system stays current and secure.
- 📦 **Portability with Containers:**
  Running Kali in a Docker container provides a portable, reproducible, and isolated environment ideal for learning and testing.

⚙️ Introduction to Docker Compose

- **What is Docker Compose?**
  A tool to define and manage multi-container Docker applications with a single YAML file.
- **Benefits:**
  - Simplifies deployment and configuration.
  - Ensures consistency across environments.
  - Makes it easy to start, stop, and update services.

📄 The docker-compose.yml File

We will use the following configuration:

```yaml
services:
  kali-linux:
    image: lscr.io/linuxserver/kali-linux:latest
    container_name: kali-linux
    security_opt:
      - seccomp:unconfined # optional for certain tests
    environment:
      - PUID=1000
      - PGID=1000
      - TZ=America/New_York
      - SUBFOLDER=/ # optional
      - "TITLE=Kali Linux" # optional
    volumes:
      - ./kali-data:/config
      - /var/run/docker.sock:/var/run/docker.sock # optional
    ports:
      - 3000:3000
    devices:
      - /dev/dri:/dev/dri # optional
    shm_size: "2gb" # optional
    restart: unless-stopped
```

📝 Explaining the Configuration

- **services:**
  Defines our containerized service.

- **kali-linux:**

  - 🖼️ **image:**
    Uses the latest Kali Linux image from LinuxServer.io.
  - 🏷️ **container_name:**
    Names the container `kali-linux` for easy identification.
  - 🔒 **security_opt:**
    - `seccomp:unconfined` relaxes some security restrictions (optional for specific tests).
  - ⚙️ **environment:**
    - `PUID` & `PGID`: Set user/group permissions.
    - `TZ`: Configures the time zone.
    - `SUBFOLDER` & `TITLE`: Optional variables for further customization.
  - 💾 **volumes:**
    - `./kali-data:/config`: Persists container data.
    - `/var/run/docker.sock:/var/run/docker.sock`: Optional; allows Docker socket access.
  - 🔌 **ports:**
    - Maps port `3000` on the host to `3000` in the container.
  - 🖥️ **devices:**
    - Grants access to `/dev/dri` for optional graphics support.
  - 💡 **shm_size:**
    - Allocates 2gb of shared memory to improve performance.

🚀 How to Run the Container

1. **Install Docker & Docker Compose:**
   Ensure both are installed on your system.
2. **Save the File:**
   Save the above content as `docker-compose.yml`.
3. **Launch the Container:**
   In your terminal, navigate to the directory and run:
   ```
   docker-compose up
   ```
4. **Access the Environment:**
   Once running, your Kali Linux container will be available (e.g., via port 3000).

📌 Key Points about Kali Linux

- 🔐 **Purpose-Built for Security:**
  Ideal for ethical hacking, penetration testing, and forensics.
- 🛠️ **Rich in Tools:**
  Provides a suite of pre-installed tools like Nmap, Wireshark, and Metasploit.
- 🎓 **Educational Value:**
  A great platform for students and professionals to learn and practice secure computing.
- 📦 **Flexible & Portable:**
  Running Kali in a container offers quick setup and easy management across various systems.

✅ Conclusion

Docker Compose streamlines the deployment of Kali Linux, making it easier to set up a secure, reproducible testing environment.
By combining containerization with a security-focused OS, you can enhance your lab sessions and provide a hands-on learning experience.

Thank you for your attention!